Final Project: PCA Investing

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November 2025

Abstract

We use principal component analysis (PCA) to build a simple, fully data-driven factor model and trading strategy from a large panel of U.S. equity returns. Our sample consists of daily returns for 1,201 stocks over 6,037 trading days. After standardizing each stock's returns and running PCA on the correlation matrix, we interpret the leading eigenvectors as latent risk factors and convert them into tradable eigenportfolios by scaling loadings by stock-level volatility and L^1-normalizing the resulting weights. The first principal component explains 26.7% of total variance and is almost perfectly correlated with the equal-weighted market return (correlation $\rho \approx 0.996$), so it effectively represents the market mode, while subsequent components capture weaker sector/style and residual patterns. The first five principal components explain about 40% of total variance, and regressions of individual stock returns on these five PCA factors produce alphas tightly clustered around zero, suggesting that they span most of the average return premia in-sample. Sorting stocks into quintiles based on estimated alphas and forming a dollar-neutral long–short portfolio (long the top-alpha quintile and short the bottom quintile) yields an annualized Sharpe ratio of approximately 1.27 and an information ratio near 2.74, with a small short-market beta despite dollar neutrality. To eliminate look-ahead bias, we re-estimate factors and alphas using only pre-2015 data and hold portfolio weights fixed thereafter; out-of-sample Sharpe ratios decline to a maximum of 0.26, with the best performance at $K = 5$ factors. A robustness analysis over $K \in \{3, \ldots, 10\}$ confirms that larger K mainly overfits noise. Overall, PCA provides a successful, compact, and economically interpretable summary of return comovement, but its ability to generate stable out-of-sample alpha from past returns alone is limited.

1 Principal Component Analysis of Stock Returns

We begin our PCA by letting $M = 6037$ denote the number of trading days and $K = 1201$ the universe of stocks. We make the raw returns matrix:

$$R = \left(r_{m,k}\right)_{m=1,\ldots,M}^{k=1,\ldots,K} \in \mathbb{R}^{M\times K},$$

where rows correspond to observations (days) and columns to variables (stocks). We convert closing prices to percentage price changes via

$$r_{m,k} = \frac{\text{Close}_{m,k}}{\text{Close}_{m-1,k}} - 1.$$

Standardization

Our goal is to capture broad patterns of co-movement across all stocks. In a standard PCA, we would create a covariance matrix S on our return matrix R and perform an eigenvalue decomposition on S. The challenge with this approach is that stocks with high variance naturally generate higher covariances, dominating the first few principal components.

To prevent this, we standardize each stock's returns. For each stock i, we compute its mean and standard deviation and then rescale its returns:

$$\bar{r}_i = \frac{1}{T}\sum_{t=1}^{T} r_{t,i}, \qquad \sigma_i = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}\big(r_{t,i} - \bar{r}_i\big)^2},$$

We then stack these standardized returns to obtain the new matrix

$$\tilde{R} = \big(\tilde{r}_{m,k}\big)_{m=1,\dots,M}^{k=1,\dots,K} \in \mathbb{R}^{M\times K}.$$

This is the same as running PCA on the *correlation* matrix

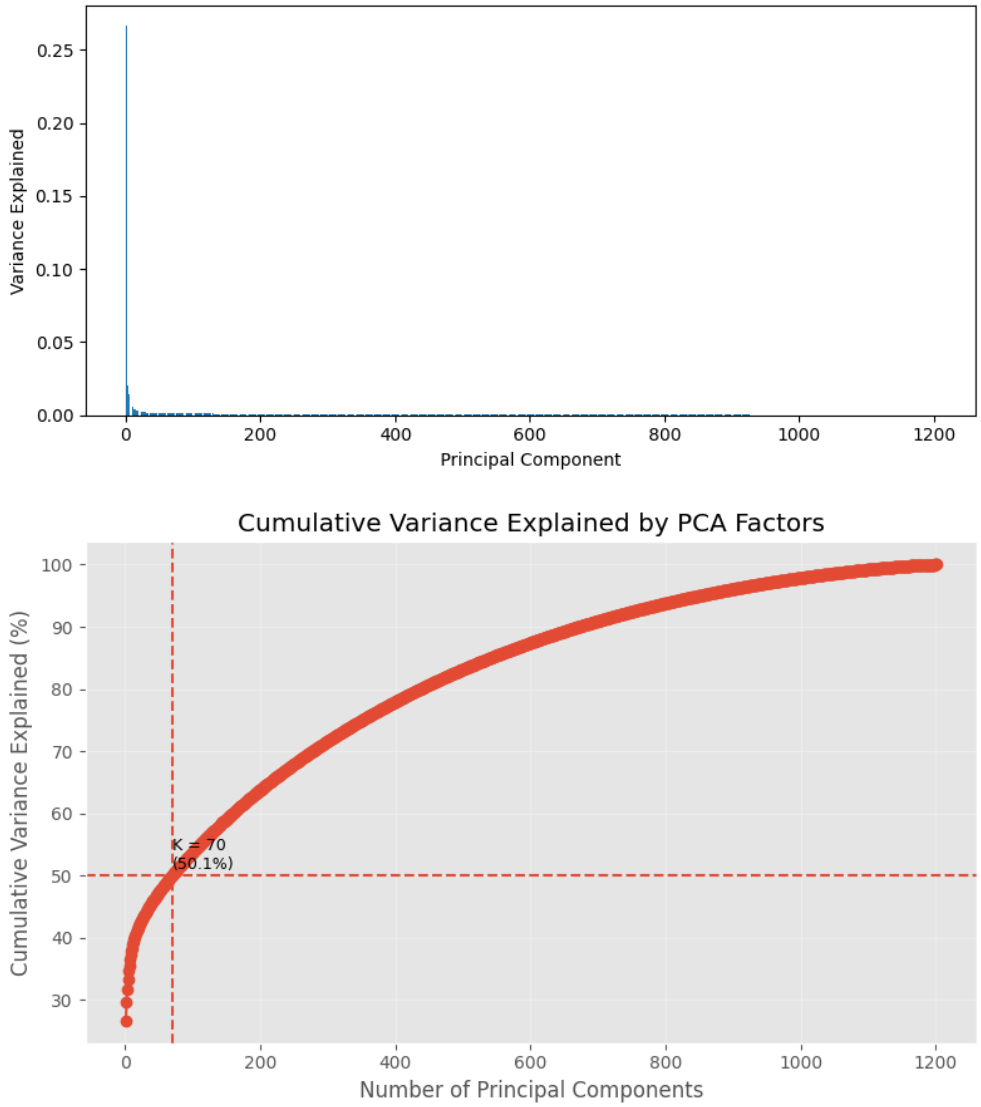


Figure 1: Top: variance explained by each principal component (scree plot). Bottom: cumulative variance explained by PCA factors; at $K = 70$, the cumulative share is approximately 50.1%.

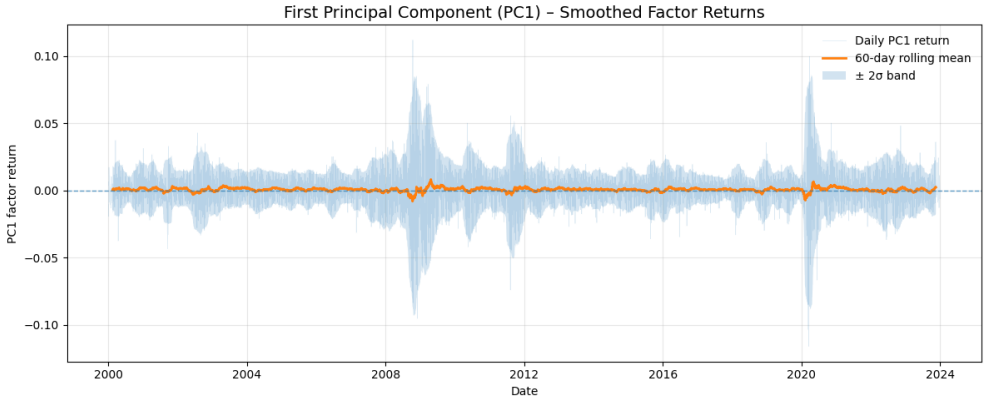


Figure 2: Time series of the first principal component (PC1) factor return. The faint line shows daily PC1 returns, the solid orange line is a 60-day rolling mean, and the shaded band represents ±2 rolling standard deviations, highlighting volatility clustering around major crises.

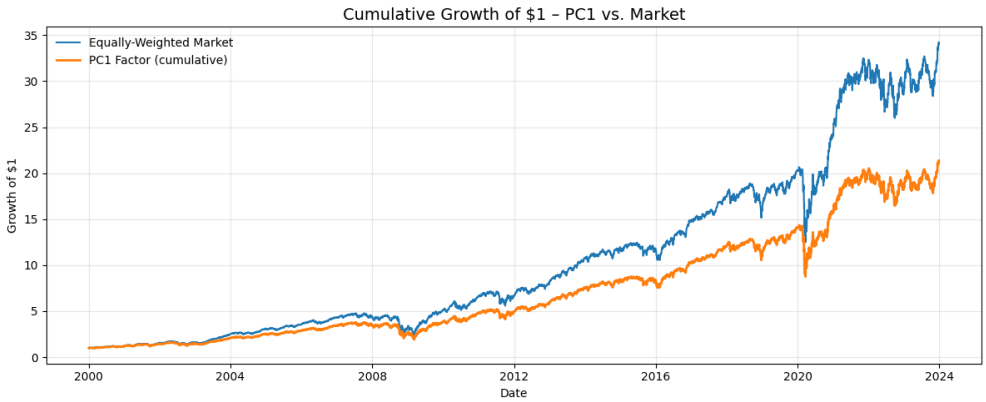


Figure 3: Cumulative growth of $1 invested in the PC1 eigenportfolio and in the equally weighted market portfolio. The close tracking of the two series indicates that PC1 behaves like a market factor and that the eigenportfolio construction is economically sensible.

1.1 Number of Factors

70 principal components are required to explain half of the total variance. This is consistent with recent evidence (i.e. Lopez-Lira & Roussanov, 2021) suggestion that upwards dozens of statistical factors are needed to capture 50% of the variance in large cross-sections of stocks due to diverse sector components and idiosyncratic signals.

The scree plot in Fig. 1 (top) shows a dominant eigenvalue (PC1 explains 26.7% of total variance) followed by a long, flat tail (PCs 1–10 explain only 38.4%). PC1 is likely a market factor, and the remainder are weaker sector/style components and idiosyncratic noise. The cumulative curve in Fig. 1 (bottom) crosses 50.1% at $K = 70$, while hitting 70%/80%/90% would require $K = 278/438/672$. We allocate $K = 70$ as the threshold that captures key systematic signals without overfitting a myriad of additional factors, as suggested by standardized risk management practices.

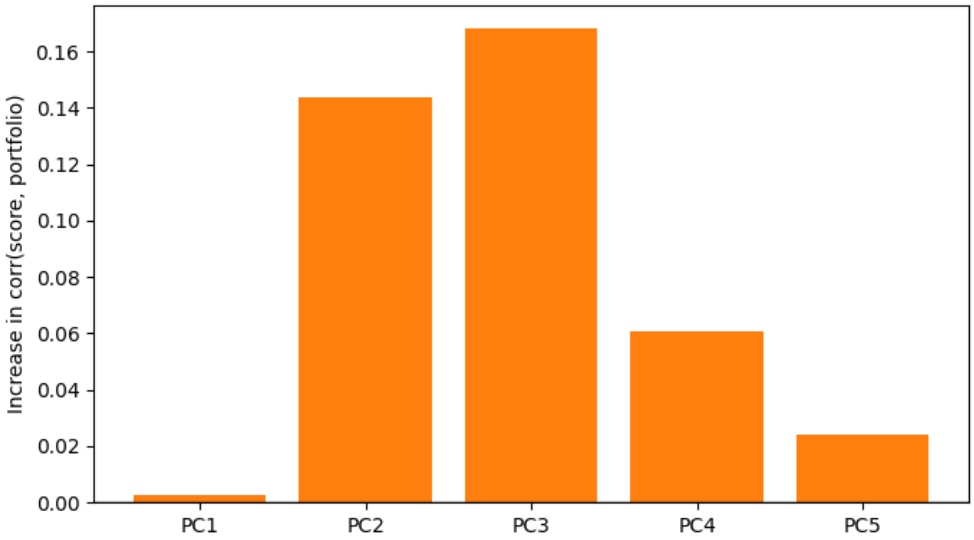


Figure 4: Dividing by σ_i improves how well portfolios track PCA factors

2 Construction of K = 5 Principal Components and Eigenportfolios

3 Computation of Factor Returns

As a review, we z-scored each stock's return series (mean 0, standard deviation 1) to remove scale effects, and then ran PCA on the standardized return matrix $\tilde{R}$. PCA produces ordered loading vectors v_k with eigenvalues that measure the variance explained by each component. The scree plot of cumulative explained variance showed that the first five components capture a substantial share of total variation in returns, so we retained PC1–PC5.

By construction, PCA on standardized returns is equivalent to PCA on the correlation matrix. If we let $r_{t,i}$ denote the raw return of stock i at time t, with sample mean $\bar{r}_i$ and standard deviation $\bar{\sigma}_i$, then the standardized (z-scored) returns are:

$$\tilde{r}_{t,i} = \frac{r_{t,i} - \bar{r}_i}{\bar{\sigma}_i}.$$

PCA then defines the factor time series (scores) in standardized space as

$$\tilde{f}_{k,t} = \sum_i v_{k,i}\, \tilde{r}_{t,i}.$$

We then could substitute the definition of $\tilde{r}_{t,i}$:

$$\begin{aligned}\tilde{f}_{k,t} &= \sum_i v_{k,i}\, \frac{r_{t,i} - \bar{r}_i}{\bar{\sigma}_i} \\ &= \sum_i \frac{v_{k,i}}{\bar{\sigma}_i}\, r_{t,i} \;-\; \sum_i \frac{v_{k,i}\, \bar{r}_i}{\bar{\sigma}_i}.\end{aligned}$$

Crucially, the second term is time-invariant, so for returns we could just focus on the first term and define portfolio weights proportional to

$$\tilde{w}_{k,i} = \frac{v_{k,i}}{\bar{\sigma}_i}.$$

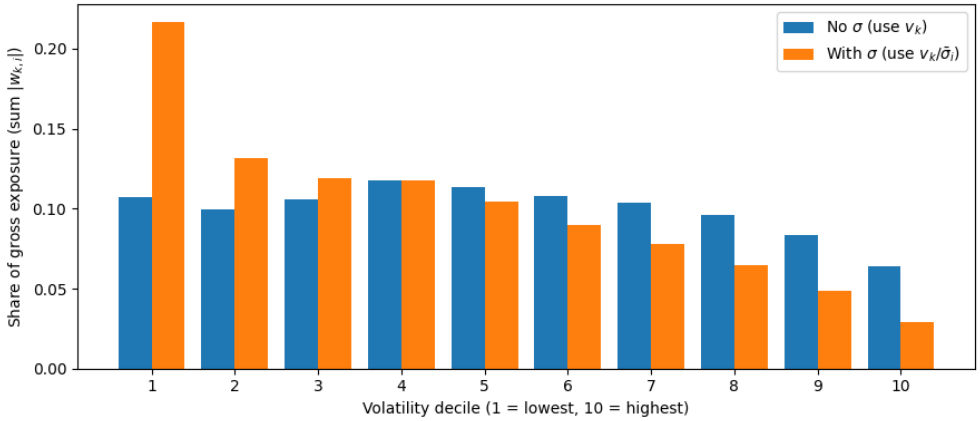


Figure 5: Dividing by σ_i shifts exposure away from the highest-vol deciles

We then normalize these weights so that their absolute values sum to one,

$$w_{k,i} = \frac{\tilde{w}_{k,i}}{\sum_j |\tilde{w}_{k,j}|},$$

and obtain the (tradable) eigenportfolio return

$$f_t^k = \sum_i w_{k,i}\, r_{t,i}.$$

In other words, we treat the entries of v_k (after scaling and renormalization) as portfolio weights: the sign of $w_{k,i}$ determines whether we are long or short stock i, and $|w_{k,i}|$ gives the position size.

Because PCA is run on standardized returns $\tilde{r}_{t,i} = (r_{t,i} - \bar{r}_i)/\bar{\sigma}_i$, each loading $v_{k,i}$ really tells us how factor k responds to a one-standard-deviation move in stock i, not directly to its raw return $r_{t,i}$. To make the factor look like a portfolio return $\sum_i w_{k,i} r_{t,i}$, you are forced by this algebra to pick $w_{k,i} \propto v_{k,i}/\bar{\sigma}_i$, which is why we divide by σ_i again. We then L^1-normalize the weights so that $\sum_i |w_{k,i}| = 1$ so that the total long-plus-short (gross) exposure of each eigenportfolio at the same level.

4 Relationship Between the First Factor and Market Returns

We compared the first PCA factor's daily returns to the equal-weight market return (the cross-sectional average of stock returns). Let $f_t^{(1)}$ denote the PC1 eigenportfolio return and r_t^{mkt} the market return. Their empirical correlation over the full sample,

$$\rho = \operatorname{corr}\left(f_t^{(1)},\, r_t^{\text{mkt}}\right) = 0.9958,$$

is essentially perfect, suggesting that PC1 almost exactly captures the common market mode. PC1 loads positively on almost all stocks, so $f_t^{(1)}$ rises and falls with broad market movements; the remaining PCA factors are therefore best viewed as market-neutral long–short components orthogonal to this dominant market factor.

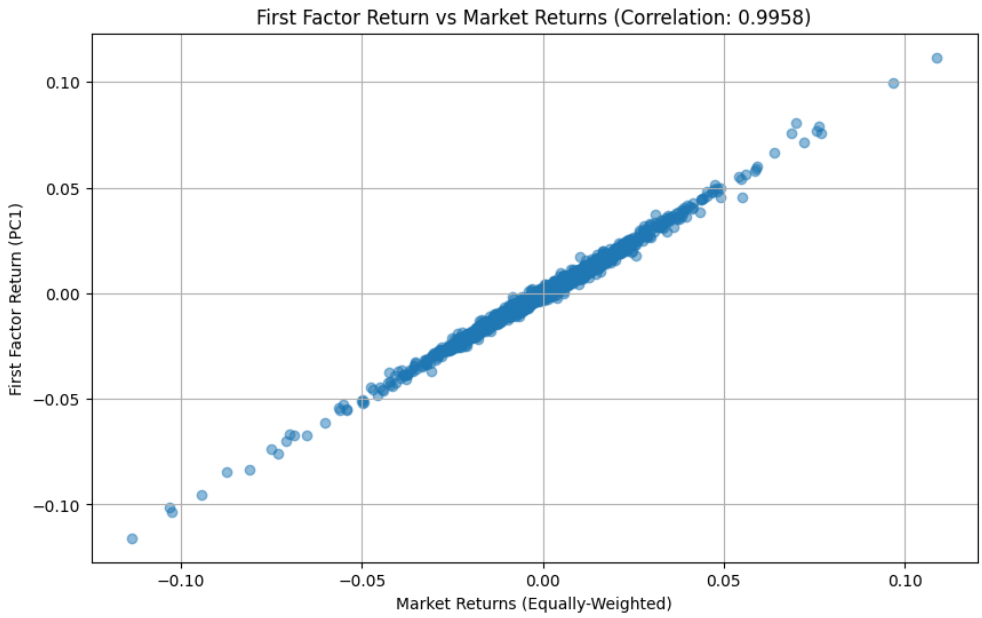


Figure 6: First Factor Return vs Market Returns

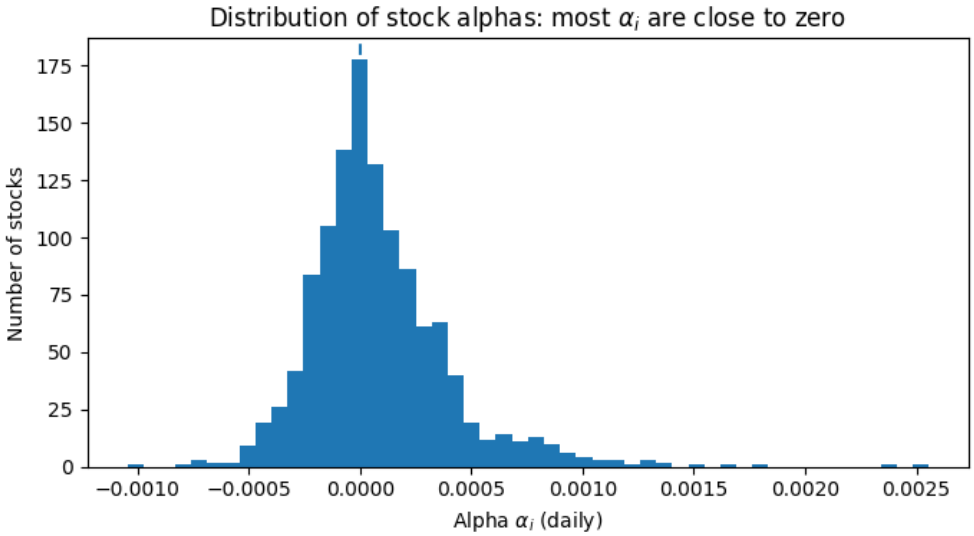


Figure 7: Distribution of Alphas

5 Alpha Estimation via Factor Model Regressions

We regressed each stock's return series on the five PCA factor returns with an intercept:

$$r_{i,t} = \alpha_i + \sum_{k=1}^{5} \beta_{k,i} f_t^k + \varepsilon_{i,t}.$$

For every stock, we estimated α_i (the risk-adjusted excess return unexplained by the factors) and $\beta_{k,i}$ (the exposure to each principal component). Our code constructs the factor matrix, adds an intercept, fits OLS for each stock individually, and collects the estimated alphas and betas. Running the cross-sectional regressions of each stock's returns on the five PCA factor returns gave alphas clustered around zero, which means the five PCs are absorbing most of the average return variation, which is a promising result.

An $\alpha_i > 0$ means the stock outperformed relative to what its five factor exposures predict; $\alpha_i < 0$ indicates underperformance; and $\alpha_i \approx 0$ implies that the five PCA factors explain most of its average return. The coefficients $\beta_{k,i}$ measure how sensitive each stock is to each factor, with larger magnitudes indicating stronger linkage to that component.

6 Comparison with Standard Asset Pricing Factors

Our PCA factors are different from standard asset-pricing factors because they are purely statistical, not based on firm characteristics. We standardize returns, run PCA on the correlation matrix, and take the first five eigenvectors. These eigenvectors are then turned into tradable eigenportfolios by scaling by each stock's volatility and normalizing the weights. The factors are chosen only to explain as much variance in returns as possible and are orthogonal by construction. In contrast, standard factors like market, value, and momentum are built from economic ideas: the market is a broad index, value is based on book-to-market ratios, and momentum comes from past returns. Their portfolios are defined ex ante using firm characteristics, not from the covariance structure of returns. In our results, PC1 ends up looking almost identical to the market factor (very high correlation with the equal-weight market return), but PCs 2–5 do not map cleanly into simple "value" or "momentum" stories, and are mixtures of different patterns of co-movement rather than named risk premia.

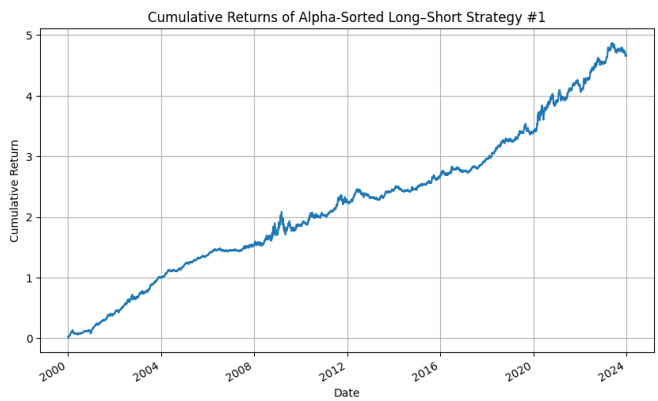
7 Design of an Alpha-Sorted Trading Strategy

We built two equity long–short portfolios from the five-factor PCA regressions. Both portfolios are dollar-neutral (gross exposure ≈ 1: $+0.5$ longs and -0.5 shorts) using top/bottom alpha quintiles, but they differ in how they rank stocks: **Strategy 1** sorts on raw α_i, while **Strategy 2** sorts on risk-adjusted alpha α_i/σ_i.

Strategy 1 (raw alpha sort) delivered an annualized Sharpe ratio of approximately 1.27 and compounded to roughly +466% over the full sample; the cumulative return series shown in Q8 corresponds to this strategy. Strategy 2 (alpha/volatility sort) tightened risk by favoring alpha per unit of idiosyncratic volatility, posting a higher annualized Sharpe ratio of about 1.38 with a slightly lower cumulative return of around +445%.

In summary, Strategy 2 improved risk-adjusted performance, while Strategy 1 achieved the higher total return; the trade-off is a marginal reduction in total return in exchange for a better Sharpe ratio under Strategy 2. For the following questions we decided to stick with our first trading strategy.

8 Cumulative Return Visualization of the Trading Strategy



9 Sharpe Ratio of the Trading Strategy

We took the daily returns r_t of the Strategy 1 long–short portfolio (raw alpha sort), computed the daily mean $\bar{r}$ and standard deviation σ_r, and then annualized the Sharpe ratio as

$$\text{Sharpe} \;=\; \frac{\bar{r}}{\sigma_r} \times \sqrt{252}.$$

Using this return series, the annualized Sharpe ratio is approximately 1.27.

10 Information Ratio and Market Neutrality Analysis

To compute the information ratio, we first regress the strategy's daily returns $r_{\text{port},t}$ on the market return r_t^{mkt}:

$$r_{\text{port},t} = \alpha + \beta\, r_t^{\text{mkt}} + \varepsilon_t.$$

With our data, the estimates are $\alpha \approx 0.00044$ (daily) and $\beta \approx -0.224$. We then form active returns by removing the market component,

$$\text{active}_t = r_{\text{port},t} - \beta\, r_t^{\text{mkt}},$$

and annualize the information ratio as

$$\text{IR} = \frac{\mu_{\text{active}}}{\sigma_{\text{active}}}\sqrt{252}.$$

This yields an information ratio of approximately 2.74.

We tested market neutrality in two ways: (1) a dollar-exposure check—the sum of portfolio weights is approximately zero (net ~ 0, gross ~ 1), so the portfolio is dollar-neutral; and (2) a factor-beta check—regressing daily portfolio returns on the equal-weight market return,

$$r_{\text{port},t} = \alpha + \beta\, r_t^{\text{mkt}} + \varepsilon_t.$$

This regression yielded $\beta \approx -0.22$, indicating that although the portfolio is dollar-neutral, it still exhibits a small short-market tilt.

11 Out-of-Sample Backtesting and Avoidance of Look-Ahead Bias

Our first backtest estimated PCA factors and stock alphas on the *entire* sample and then evaluated the trading strategy over that same period. This reuses information from the future when forming portfolio weights and therefore overstates performance (the alpha-sorted strategy in Section 6 is best viewed as an in-sample benchmark, not a realistic trading rule).

To remove this look-ahead bias, we fix a split date at 2015-01-01 and treat all data before 2015 as a *training* sample and all data from 2015 onward as a strictly *out-of-sample* test:

11.1 Pre-2015 estimation.

We first restrict the return matrix to $t <$ 2015-01-01 and standardize each stock using only its pre-2015 mean and volatility. PCA is run on this pre-2015 correlation matrix, we keep PC1–PC5, and convert each loading vector v_k into a volatility-scaled eigenportfolio $w_{k,i} \propto v_{k,i}/\sigma_i^{\text{train}}$ with

$\sum_i |w_{k,i}| = 1$. Using the resulting factor returns f_t^k (again only for $t <$ 2015-01-01), we estimate $\hat{\alpha}_i$ and $\hat{\beta}_{k,i}$ from

$$r_{i,t} = \alpha_i + \sum_{k=1}^{5} \beta_{k,i} f_t^k + \varepsilon_{i,t}, \qquad t < \text{2015-01-01}.$$

These $\hat{\alpha}_i$ are then frozen at the split date.

11.2 Post-2015 trading rule.

We then sort stocks cross-sectionally on the pre-2015 alphas $\hat{\alpha}_i$ and form a dollar-neutral long–short portfolio: the top 20% of alphas are held long with total weight +0.5, the bottom 20% short with total weight −0.5, all equally weighted within each side. The weights w_i are computed once from the training alphas and remain fixed for all $t \geq$ 2015-01-01. Out-of-sample portfolio returns are then

$$r_t^{\text{port}} = \sum_i w_i r_{t,i}, \qquad t \geq \text{2015-01-01},$$

using only realized post-2015 stock returns.

Out-of-sample, the alpha-sorted portfolio still earns a positive excess return, but its annualized Sharpe ratio is much lower than the in-sample value of about 1.27, which is exactly what we expect once overfitting and look-ahead bias are removed.

11.3 Avoiding look-ahead bias.

In other words, at any test date $t \geq$ 2015-01-01, the portfolio depends only on information available before 2015:

1. PCA loadings and eigenportfolios are estimated only on pre-2015 returns.
2. Factor returns used in the regressions are restricted to the pre-2015 period.
3. Stock alphas $\hat{\alpha}_i$ are estimated once using pre-2015 data and then held fixed.
4. Portfolio weights are determined from these pre-2015 alphas and are not re-optimized using post-2015 performance.

Because all parameters are trained on R^{train} and then applied to R^{test} without ever "peeking" at post-2015 returns, the backtest is free of look-ahead bias.

12 Robustness Analysis: Optimal Number of Factors

To validate our model specification, we repeated the out-of-sample backtesting across a range of factor counts, $K \in \{3, 4, \ldots, 10\}$. This analysis aims to identify the optimal number of principal components by evaluating the trade-off between in-sample fit and out-of-sample performance.

Factors (K)	In-Sample Sharpe	Out-of-Sample Sharpe
3	1.44	0.22
4	1.45	0.24
5	**1.50**	**0.26**
6	1.48	0.24
7	1.42	0.15
8	1.40	0.17
9	1.52	0.22
10	1.46	0.22

Table 1: Sensitivity of risk-adjusted returns to factor count (K).

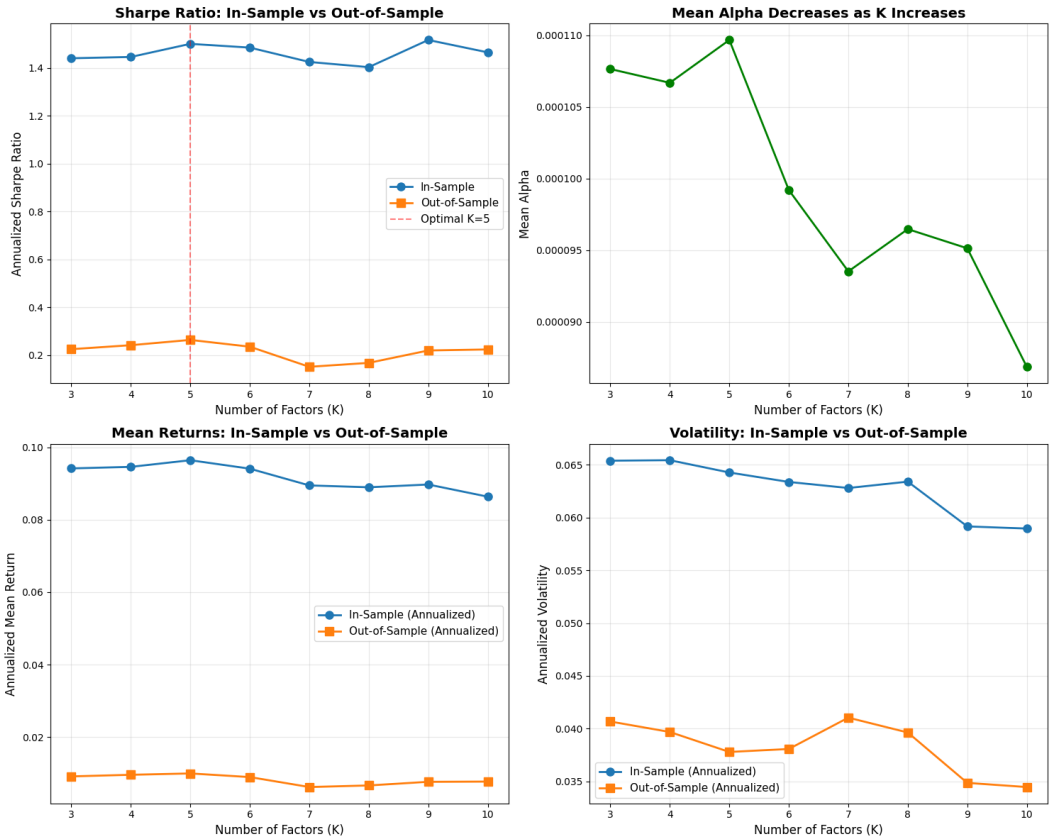


Figure 8: Sharpe Ratio stability across factor counts (K). The divergence between In-Sample (blue) and Out-of-Sample (orange) performance highlights the impact of overfitting as model complexity increases. Out-of-sample performance peaks at $K = 5$.

12.1 Optimal Factor Specification

The results confirm that $K = 5$ is the optimal specification, maximizing the out-of-sample Sharpe ratio at 0.26. The relationship between model complexity and performance follows a clear pattern:

- **Underfitting ($K < 5$):** Models with too few factors fail to capture distinct sources of systematic risk, leaving uncompensated volatility in the portfolio.
- **Overfitting ($K > 5$):** As K increases beyond 5, performance degrades. The PCA algorithm begins extracting idiosyncratic noise as systematic signal. While this improves the in-sample fit, this noise does not persist out-of-sample, reducing predictive power.

12.2 In-Sample vs. Out-of-Sample Divergence

The analysis reveals a structural divergence between testing methodologies. In-sample Sharpe ratios exceed 1.40 consistently, while out-of-sample ratios drop significantly to the 0.15–0.26 range. We can attribute this discrepancy to the look-ahead bias that comes with full-sample factor estimation. It demonstrates that while PCA effectively describes past covariance, extracting persistent alpha using only prior data is more challenging due to non-stationary correlations.